

Towards a Unified Definition of Social XR

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ABSTRACT

Social Extended Reality (Social XR) is an evolving domain within Extended Reality (XR), encompassing immersive technologies such as Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR). Despite its growing significance in both research and practice, the term “Social XR” remains ambiguously defined, with inconsistent usage across academic and industry contexts. This paper seeks to address this challenge by synthesising existing definitions and interpretations of Social XR, aiming to encourage a clearer, unified conceptualisation. Through an analysis of current research, we identify key themes underpinning the concept of Social XR, such as shared immersive 3D environments, blending of virtual and physical elements, real-time multiuser interaction, and user representation. Thus, we propose a comprehensive definition of Social XR, aiming to stimulate further discussion, encouraging refinement toward a coherent understanding of the term. Additionally, we build on previous work and introduce the XR3C Model, for a better understanding of collaboration within immersive environments and its relation to Social XR.

Index Terms: Extended reality (XR), immersive experience, social computing, multimedia communication, collaboration

1 INTRODUCTION

Social Extended Reality (Social XR), a domain within Extended Reality (XR), has become an emerging topic in both academic research [5] and practical applications, as it explores the potential for real-time social communication in immersive environments. As XR technologies such as Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR) continue to evolve, they offer new opportunities for users to engage in shared three-dimensional (3D) spaces that blend the digital and physical worlds. Social XR, which allows users to interact and communicate within these spaces, has gained increasing attention over the years [33], driven by the growing importance of social platforms and virtual collaboration tools [23, 34]. However, despite its relevance, there remains a significant ambiguity in how the term “Social XR” is defined and understood, with various interpretations in both academia and industry. This inconsistency in terminology complicates efforts to establish clear guidelines, making it difficult to compare studies and develop unified frameworks for Social XR.

With these challenges in mind, this paper contributes by: 1) providing an overview of existing definitions and interpretations of “Social XR” and related terms in the academic literature, 2) synthesising a definition of Social XR that combines identified key elements to facilitate a unified understanding of the term, and 3) introducing the novel XR3C Model, to address the need for a deeper understanding of collaboration in XR and, by extension, Social XR.

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2 DEFINITION OF SOCIAL XR

2.1 XR Terminology

Understanding the evolving landscape of immersive technologies requires distinguishing between AR, VR, and MR – often collectively referred to as XR – grounded in established standards and recent publications. According to [17], based on definitions from [28], **VR** is defined as “an environment fully generated by digital means”, where the virtual environment is different from the local physical environment. **AR** is characterised as “an environment containing both real and virtual sensory components”, contextually placing digital content in the physical environment in real time [28]. To capture the variations within AR, the concept of AR continuum is introduced [17], ranging from “virtual content clearly overlaid on a real environment” (Assisted Reality) to “virtual content that is seamlessly integrated and interacts with a real environment”, defining Mixed Reality. **MR** therefore adds to the concept of AR by creating environments “containing both real and virtual components that are seamlessly integrated and interact with each other in a natural way”, [17] merging both into a hybrid space where physical and digital objects coexist and interact in real time with high levels of local presence [28].

The broader term Extended Reality (XR), as per [28] and adopted by [17], refers to “an environment containing real or virtual components or a combination thereof”, with the “X” serving as a placeholder for a new environment (e.g., “V” for VR). However, other interpretations, such as those in [1] and [2], define XR as an umbrella term that encompasses the entire continuum of “real-and-virtual combined environments.” The XR continuum in question integrates VR, AR, and MR to achieve user immersion through human-computer interaction and various multisensory devices.

2.2 Existing Social XR Definitions

The term “**Social XR**” and its derivatives such as “**Social VR/AR/MR**” remain in an evolutionary stage, with definitions that are neither fully established nor universally agreed upon amongst researchers. This ambiguity reflects a broader challenge in defining the scope of XR, as discussed in Section 2.1, leading to interchangeable usage of related terms, further complicating efforts to establish standardised terminology.

Lounakoski et al. [20] highlight that XR research has focused primarily on experiences designed for a single user, with limited focus on multiuser scenarios. They further emphasise that Social XR research remains fragmented, predominantly addressing Social VR [32], with relatively little exploration of Social AR and MR. Consequently, the evolving terminology and relatively nascent state of Social XR research have resulted in limited literature that explicitly addresses variations of the term. However, research addressing related phenomena may be discussed under alternative terminologies, including keywords such as “shared,” “collaborative,” or “metaverse”, often combined with XR-related terms (X|V|A|M)R. This research mainly explores the shared social experiences of multiple users that involve varying levels of computer-aided augmentation of social interactions in fully or partially virtual environments, making it suitable for association with the term “Social XR”.

Table 1: Definitions and Interpretations of the Term “Social XR” and Related Terminology from Various Publications

Reference	Definition
Billinghurst et al. (2024) [5]	“ Social Extended Reality (XR) systems promise to overcome the limitations of current real-time teleconferencing systems, enabling a better sense of immersion, enhancing the sense of presence, and fostering more successful interpersonal interactions.” “With Social Extended Reality (XR) emerging as a new medium, where users can remotely experience immersive content with others, the vision of a true feeling of ‘being there together’ has become a realistic goal.”
Cortés et al. (2024) [8]	“To overcome the limitations of 2D videoconferencing, Social eXtended Reality (XR) has emerged as a promising solution by offering a more natural and immersive communication. This is because of the inherent 3D nature of XR technology, which allows users to freely move around and interact with each other in a way that is more realistic and engaging than ever before. In addition, under the XR paradigm, local and distant physical realities can be blended with virtual assets to offer realistic interactions in 6 Degrees of Freedom (DoF) that enhance the user experience. Within the possibilities offered by this paradigm, Social XR communications are called to be the next step in immersive communications.”
El Saddik et al. (2024) [10]	“eXtended meta/uni/Verse (XV), also known as “Mersivity,” “Vironmentalism,” “ Social XR ,” “Physical metaverse,” etc., combines eXtended Reality (XR), eXtended Intelligence (XI), eXtended Being (XB), including Digital Twin, eXtended Economy (XE), and eXtended Society (XS) into a vision that covers a generally agreed metaverse scope while extending the impacts to include physical and social implications. XV is the next evolution at the nexus of three axes: the 1) physical; 2) virtual; and 3) social.”
3rd Generation Partnership Project (3GPP) (2023) [1]	“ Social XR is used as an umbrella term for combining, delivering, decoding and rendering XR objects (avatars, conversational, sound sources, streaming live content, etc.) originating from different sources into a single user experience. Social XR may be VR centric, but also may apply to AR and MR. Social XR is expected to integrate multiple XR functionalities such as 6DoF streaming with XR conversational services.”
Cortés et al. (2023) [7]	“ The social extended reality (social XR) is conceived as a shared space, where users can interact with others in an immersive environment that mixes elements from the physical reality with the virtual one. Thus, Social XR requires an immersive interactive communication.” “The Social XR is intrinsically intertwined with social interaction. Therefore, any analysis of the use cases pertaining to the Social XR ought to focus on scenarios, where users are transported to remote environments and engage in social interactions. Consequently, the Social XR must feature a communication system, which relies on immersive communication. In addition, immersive communication necessitates the use of devices such as head-mounted displays (HMDs), coupled with the processing of an interactive shared world. To facilitate interaction within this shared world, audiovisual and haptic techniques are indispensable. All these functionalities are combined through different processes to generate the immersive environment.”
Krome and Kopp (2023) [18]	“Virtual worlds, on the other hand, provide interactive environments that enable users to dive into alternate realities together with another. The use of such worlds for social gatherings can be summarised under the umbrella term “ Social eXtended Reality ”, or “ Social XR ”. This encompasses the use of Virtual Reality, Augmented Reality as well as regular display devices to engage with this social space.” “Social XR varies a lot, mainly in terms of immersion and presence. The degree of immersion heavily depends on the sense of agency over ones avatar as well as how realistically other avatars look and behave, and impacts the resulting behavior of users”
Mann et al. (2023) [22]	“We propose eXtended meta-uni-omni-Verse (XV) as an overarching term, concept, and taxonomy for shared (social) XR across all of the “verses,” including the universe (physical reality, i.e., atoms), the metaverse (virtuality, i.e., bits), the omniverse and multiverse, etc.” “We propose XV for shared, social, collaborative XR, i.e., to take the space of all realities in and beyond the a-b plane and extend them into a third, sociality dimension, ... a 3-D nature-technology-human taxonomy volume.”
Görllich (2022) [14]	“ Social XR allows several people or a group of people to experience social presence and co-presence by engaging in activities and real-time conversations with each other.”
Pérez et al. (2022) [26]	“ Social Extended Reality is a system archetype that addresses both visual communication and shared immersion.” “Differing from the given examples of pure visual communication, here the use of a Virtual Reality or Augmented Reality HMDs are required. Also, unlike the given examples of pure shared immersion, the realism of users is important.” “It is the integration of real-time capture of a person, typically using volumetric video systems, into a shared virtual environment.” “... users are represented using 3D point clouds obtained from volumetric video captures.” “The components involved in Social XR are similar to those present in Shared XR, but with some subtle differences: HMD displays tend to be more AR or XR goggles, avatars are based on video or volumetric capture to increase their realism; worlds might either be CGI scene or scenes based on photogrammetry, or none.” “All type of actions related to object manipulation, pointing and locomotion are supported and referred to the virtual world.”
Stapleton and Megale (2021) [30]	“ Social eXtended Reality (SocialXR) integrates spatial computing (virtuality) with the mapping of human relationships and sociality using perceptual computing.”
Hilken et al. (2020) [15]	“We identify this new class of AR as social AR , which we define as a technology that enables two or more users to communicate by sharing and virtually enhancing a common view of the physical environment.”
Hirskyj-Douglas et al. (2020) [16]	“The use of augmented reality technology to initiate, support, encourage or mediate in-person interactions of two or more people. The augmentation can involve digitally adding or transforming audiovisual input.” “... which we define as the interactive and social experiences of two or more co-located people enhanced by digital information. Social AR includes creating or sharing information, ideas, interests, and other forms of expression with digital content overlaid onto the physical world. We are particularly interested in how these digital augmentations shape and affect people’s face-to-face interactions.”
Maloney et al. (2020) [21]	“ Social VR refers to 3D virtual spaces where multiple users can interact with one another through VR head-mounted displays (HMDs).”
Perry (2015) [27]	“At its most basic, social virtual reality allows two geographically separated people, in the form of fairly realistic avatars, to communicate as if they were face-to-face. They can make eye contact and can manipulate virtual objects that they both can see.”

Although incorporating studies with related concepts under different terminologies would be ideal, the sheer breadth and diversity of such research hinder a comprehensive analysis of Social XR. Therefore, the analysis in this paper focusses on bodies of work that explicitly use the term “Social XR” or its derivatives, where “X” represents a specific technology (i.e. V for VR, A for AR, M for MR), or the umbrella term “XR”, as previously discussed in Section 2.1. The goal is to examine the outcomes of efforts aimed at defining, interpreting, and adopting these terms in research, exploring their coherence and consistency. Furthermore, our objective is to synthesise these existing definitions of “Social XR” by identifying salient characteristics to propose a comprehensive and unified interpretation of the term.

Despite the fact that there are various research papers that make use of Social XR-related terms, few provide explicit definitions or detailed explanations of these terms or even refer to established definitions from prior research. In several instances, these terms appear only in the title or abstract, with no further elaboration in the rest of the work. For example, [11, 31] use the term in the title, but do not define or explore it beyond the abstract. This leaves the interpretation of these terms largely to the reader. Studies of this nature were thus excluded from further analysis, as our aim was to review and analyse the definitions and interpretations of Social XR.

As a result, Table 1 compiles several existing definitions and interpretations of the term “Social XR” and related terms from the relevant literature. The following Section highlights the key features of Social XR that emerge from these interpretations, while also examining the diversity in conceptualisations and the persistent challenges in achieving terminological consistency.

2.3 Discussion on Definitions and Interpretations

Based on an analysis of the definitions and descriptions of Social XR and related concepts given in Table 1, we conclude that the literature reveals a broad and evolving concept that encompasses various dimensions of shared interaction and immersion. Central to most descriptions is the emphasis on shared 3D spaces where multiple users simultaneously engage in real-time interactions, whether in fully virtual (computer-generated or photogrammetric) scenes or virtually enhanced physical environments. Real-time interaction is thus noted as essential for effective communication within shared spaces, consistently reflected in works such as [26, 7, 15, 18, 21].

Several definitions, including [27, 15, 26], highlight the role of shared interaction. This also involves the users’ ability to mutually manipulate virtual objects within immersive environments. The enhanced sense of immersion is another recurring element, with [5, 8] stressing how Social XR creates more engaging and realistic experiences than traditional 2D tele- and videoconferencing.

Notably, perspectives on the scope of Social XR vary. While Mann et al. [22] and El Saddik et al. [10] position Social XR within broader frameworks such as eXtended meta/uni/(omni)/Verse, which encompasses three dimensions – physicality, virtuality, sociality – others, such as [30], adopt a narrower focus, describing Social XR as the integration of virtuality and sociality.

Discrepancies also arise in whether co-located or remote interactions are supported, characterising what a shared environment is for a specific Social XR experience. Although [16, 15, 27] all emphasise face-to-face communication, their focus differs: Hilken et al. [15] and Hirschy et al. [16] focus on co-located (in person) users interacting through AR, while Perry [27] addresses remote (geographically separated) users in VR. Billinghurst et al. [5] and Cortes et al. [8] also recognise the role of remote communication within Social XR environments, with [8] specifically noting the potential for both remote and co-located interactions.

Furthermore, Pérez et al. [26] refer to the blending of physical and virtual elements in Social XR as “shared immersion.” This is further elaborated in [7, 8], which illustrate how, through blending,

Social XR facilitates immersive communication and shared experiences. Moreover, [5, 14, 18] highlight how the blending of physical and virtual enhances presence through increased immersion.

Outlooks on Social XR also differ in terms of the number of users involved in the experience and their representation within the environment. For instance, [15] and [16] describe AR-based social interactions as experiences involving two or more users sharing a common physical environment, particularly in co-located settings. In contrast, [27] focusses on scenarios solely with two users in VR, while [21] expands this to multiple users within VR environments. The representation of these users further varies – some publications, such as [26], advocate for highly realistic 3D point clouds from volumetric video, whereas others, such as [1] and [27], emphasise avatars with varying levels of realism.

2.4 Additional Considerations

While Rauschnabel et al. [28] do not directly reference “Social XR” and are therefore excluded from Table 1, their work offers valuable insights into how AR and VR facilitate shared and social experiences, albeit in distinct ways. It indicates that AR enables multiple users to interact with shared virtual content in co-located physical environments, leveraging virtual object persistence to foster co-presence and both human-system and human-human interactions. In contrast, VR is said to include users represented by avatars in fully virtual environments, typically separated physically. VR social interactions are then mediated through avatars, relying on realistic representations and careful management of non-verbal cues and spatial dynamics to simulate authentic social exchanges, suggesting that users in VR social experiences are physically apart. This comparison highlights focus on shared physical-world experiences in AR versus the facilitation of remote interactions in virtual spaces within VR. However, recent advancements have blurred the lines between AR and VR, with modern head-mounted displays enabling seamless engagement between the two. Moreover, not only can a single device support and switch between multiple environments [3], but a single service can also accommodate users operating in different environments simultaneously [6]. This convergence fosters flexible immersive interactions, allowing users to engage in simultaneous heterogeneous XR experiences. Consequently, hybrid experiences should be considered within the Social XR context, as they expand possibilities for interaction, immersion, and presence, while also introducing new complexities.

Social XR can encompass diverse user interactions, such as pure communication without task-oriented goals or competitive activities. However, collaboration emerges as a particularly prominent application of Social XR. Albeit not all Social XR use cases are inherently collaborative, the technology is increasingly leveraged as a tool for collaboration [33, 34], suggesting the need to consider it in the conceptualisation of Social XR. Although an in-depth overview of each aspect is essential, our focus shifts toward the collaborative potential. While most interpretations compiled in Table 1 emphasise communicative and interactive experiences in immersive environments, explicit references to collaboration are notably sparse. Exceptions include mentions of collaborative XR [22] and “creating or sharing information” [16]. Two other publications [8, 30] do not directly address collaboration in their Social XR definitions, but the user tasks in respective studies were described as collaborative. Beyond these examples, several other publications outside this analysis showcase the application of Social XR in collaborative scenarios without explicitly defining or interpreting the term “Social XR” [24, 25]. Given this, collaborative possibilities emerge as an important area of application within Social XR.

2.5 Synthesised Definition of Social XR

The overall analysis, drawing from a range of definitions and descriptions, illustrates the multifaceted nature of Social XR and its

evolving conceptualisation within the literature. Although its full scope remains in development, several elements emerge that could form the basis of a unified definition. General ideas revolve around shared 3D environments, whether fully virtual or a blend of virtual and physical, where users can participate in real-time social interactions and communication, enhancing presence and immersion. Additionally, the ability to manipulate virtual objects within these shared spaces and the varying degrees of user representation, from avatars to volumetric models, are key aspects that surface across definitions. Thus, based on these recurring themes in the examined literature, we propose the following broad definition of Social XR:

Social XR can be defined as an immersive multiuser experience that involves real-time social interaction and communication within a shared virtual or augmented space. It supports co-located or remote users, aiming to enhance presence and may foster engagement through interactive environments where users can manipulate physical or virtual objects and can take on diverse forms of user representation.

3 XR3C COLLABORATION MODEL

Given the lack of explicit collaborative focus in explored Social XR interpretations in Section 2.3, despite the abundance of collaborative use cases, a structured framework is needed to bridge this gap. Since elements of Social XR interpretations could be tied to existing principles of collaboration, analysing the concept of collaboration in XR becomes vital for not only obtaining a better grasp of Social XR, but also a better overview of collaboration in XR in general.

The “social” aspect of Social XR is rooted in social interactions [9], a concept deeply related to human behaviour. Within XR, these interactions evolve to encompass unique dimensions shaped by immersive environments [4]. Communication, [12], a core example of social interactions, similarly adapts to XR, reshaping traditional concepts to address immersive spaces [29]. Collaboration, one of the key manifestations of human communication [19], has thus also naturally progressed from real-world practices to digital realms.

Fuks et al. [13] introduced the **3C Collaboration Model** in the context of groupware, outlining the model of computer-supported user collaboration, while emphasising the interconnected relationship between *Communication*, *Coordination*, and *Cooperation*. According to [13], the three Cs are described as follows:

- **Communication:** Exchange of information and negotiation of decisions between group members.
- **Coordination:** Management of communication, interdependencies between tasks, and resource allocation to ensure effective collaboration between group members.
- **Cooperation:** Collective task execution, as well as creation and manipulation of shared resources within a collaborative workspace shared by the group members.

The 3C Model depicts collaboration as a cyclical and interconnected process, where iterative communication and coordination are essential to address emerging challenges and adjust tasks as needed to achieve effective cooperation.

While originally developed for traditional software tools, the 3C Model can be adapted to modern immersive environments such as XR by leveraging advanced technologies to enrich traditional principles [29], fostering more interactive and engaging collaborative experiences. Such systems can furthermore enhance collaboration by facilitating communication through verbal and non-verbal interactions, enabling coordination via real-time task management and resource alignment, and fostering cooperation through collective engagement with physical and virtual content, all within shared immersive spaces. To address the need for a structured framework in XR collaboration, we propose an expansion of the traditional 3C

Model to XR contexts, referred to as **XR3C Collaboration Model**. We propose the XR3C Model to include the following:

- **Communication:** Real-time verbal and non-verbal information exchange (e.g., gestures, facial expressions), fostering dynamic social interactions within a shared immersive multiuser environment.
- **Coordination:** Communication management in a shared immersive multiuser environment, supported by interactive aids for task alignment, resource allocation and interdependence.
- **Cooperation:** Collective task execution, with active engagement in interactive creation and manipulation of physical and virtual content in a shared immersive multiuser environment.

Building on previously discussed XR characteristics, and incorporating identified key features of Social XR, the proposed XR3C Model refines the 3C Model for collaborative XR, acting as a stepping stone for the evaluation of current and future immersive multiuser scenarios from a collaborative perspective.

4 CONCLUSION

The concept of Social XR represents an intersection of technology and human interaction, leveraging the capabilities of XR to create immersive shared spaces for real-time social interaction and communication. This paper explored the evolving definitions and interpretations of the term “Social XR”, revealing the diversity and ambiguity present in academic discourse. Despite significant advances, challenges persist in achieving a unified definition, as the term “Social XR” remains inconsistently applied across publications. In many cases, the term is often used interchangeably with specific forms such as Social VR, AR, and MR, or other related concepts, hindering the standardisation of research methodologies and design principles, limiting the ability to draw meaningful comparisons between studies and formulate comprehensive frameworks. To address the aforementioned challenges related to Social XR, by analysing Social XR-related terms in existing literature, we identified key elements that characterise Social XR. As a result, based on related work, we proposed a comprehensive definition of Social XR which we hope will trigger further discussion within the research community to refine it as needed and reach a unified definition. Additionally, to better understand the frequent manifestations of Social XR through the lens of collaboration, we introduced the XR3C Model, an extension of the traditional 3C Collaboration Model, incorporating the unique affordances of XR technologies to offer a framework to examine collaboration and the dynamic interplay between social interaction and immersive environments.

Looking ahead, the adoption of a unified definition of Social XR has far-reaching implications for research and development in this field. A clearer conceptual foundation can support the advancement of research and design of more effective applications, foster interdisciplinary collaboration, and guide future studies to address gaps in understanding. As XR technologies continue to advance, Social XR holds the potential to redefine how people connect, communicate, and engage in both personal and professional contexts, paving the way for a new era of interactive and immersive experiences.

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